Types of Machine Learning Problems

A beginner's map — what each type predicts, an everyday analogy, and real examples

MACHINE LEARNING

1 · SUPERVISED LEARNING — has an ANSWER KEY

Learning from labeled examples — the data comes with the right answers.

Classification

Predicts	a category / label
Analogy	sorting mail into bins
Examples	spam / not-spam, disease / no-disease, churn yes / no

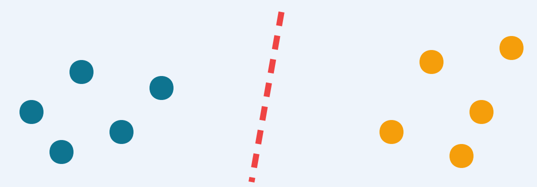
Answer looks like
one of a few fixed options

Regression

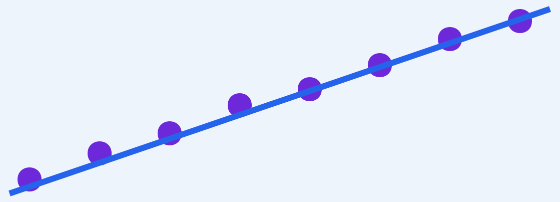
Predicts	a number on a scale
Analogy	reading a value off a dial
Examples	house price, temperature, sales, delivery ETA

Answer looks like
any value in a range

Classification — splitting categories



Regression — predicting a number



2 · UNSUPERVISED LEARNING — NO answer key

Finding patterns in data with no labels — you discover the structure yourself.

Clustering
THE MAIN ONE

What it does
groups similar items, with no labels

Analogy
sorting a photo library into look-alike piles

Examples
customer segments, grouping articles

Dimensionality reduction

What it does
compresses many features into a few

Analogy
a 50-question survey summed into 3 themes

Examples
visualization, compression, denoising

Anomaly detection

What it does
spots the rare / odd item out

Analogy
the one fake note in a stack of real ones

Examples
fraud detection, defect spotting

Good to know

- In unsupervised learning there is no "right answer" to check against — the model finds structure on its own.
- Clustering is the workhorse you will use most; the other two support and clean up your data.
- A third ML family, Reinforcement learning (learning from rewards), exists but is out of scope for this course.

Quick comparison

SUPERVISED — has answers
predict a CATEGORY = Classification · predict a NUMBER = Regression
you learn from labeled examples

UNSUPERVISED — no answers
GROUP = Clustering · SIMPLIFY = Dimensionality reduction · FLAG ODD = Anomaly detection
you discover the structure yourself